Signals and Systems Lecture-16: z-Transform 1

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Introduction (1)

- Consider a **causal** continuous-time signal $x(t)$.

$$x(t) = 0 \ \text{ for } \ t < 0$$

- The sampled version of $x(t)$ is given by:

$$x_s(t) = \sum_{n \in \mathbb{N}} x[n]\delta(t - nT_s)$$

where:
  - $x[n] = x(nT_s)$ is the n-th sample of $x(t)$.
  - T_s is the sampling period.
- Given that the Laplace transform of the unit impulse is given by:

$$\delta(t) \leftrightarrow 1 \quad ; \quad \delta(t - nT_s) \leftrightarrow e^{-snT_s}$$

- Then the Laplace transform of $x_s(t)$ is given by:

$$X_s(s) = \sum_{n \in \mathbb{N}} x[n]e^{-snT_s}$$

Introduction (2)

- We define the z-transform of the discrete time signal $\{x[n]\ ;\ n \in \mathbb{N}\} = \{x[0], x[1], x[2], \ldots\}$ by:

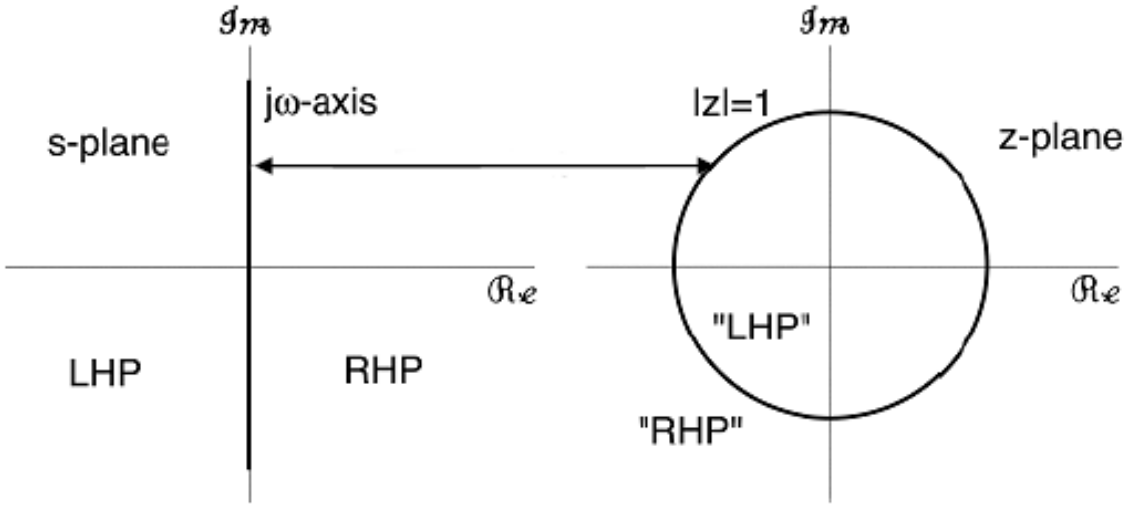
$$X(z) = \sum_{n \in \mathbb{N}} x[n] z^{-n}$$

where the complex variable z is defined by:

$$z = e^{sT_s} = e^{\sigma T_s} e^{j\omega T_s}$$

- The relation $z = e^{sT_s}$ maps points in the s-plane into points in the z-plane.

Introduction (3)



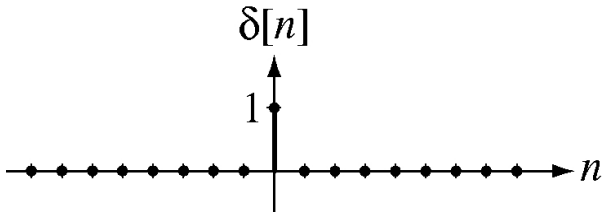
$$z = e^{\sigma T_s} e^{j\omega T_s} \Rightarrow |z| = \left|e^{\sigma T_s}\right| \left|e^{j\omega T_s}\right| = e^{\sigma T_s}$$

- The $j\omega$ axis (whose equation is $\sigma = 0$) in the s-plane is mapped into the unit circle $|z| = e^{0T_s} = 1$ in the z-plane.
- The open LHP $\sigma < 0$ is mapped inside the unit circle since $|z| = e^{\sigma T_s} < 1$ for $\sigma < 0$.
- The open RHP $\sigma > 0$ is mapped outside the unit circle since $|z| = e^{\sigma T_s} > 1$ for $\sigma > 0$.

Discrete-Time singularity functions (1)

- The Discrete-Time (DT) unit impulse is defined by:

$$\delta[n] = \begin{cases} 1, & n = 0; \\ 0, & n \neq 0. \end{cases}$$



- The z-transform of $x[n] = \delta[n]$ is given by:

$$X(z) = \sum_{n \in \mathbb{N}} x[n] z^{-n} = x[0] z^{-0} = x[0] = 1$$

- Let $k \in \mathbb{N}$ be a constant.
  - The shifted version of the DT unit impulse is defined by:

$$\delta[n-k] = \begin{cases} 1, & n = k; \\ 0, & n \neq k. \end{cases}$$

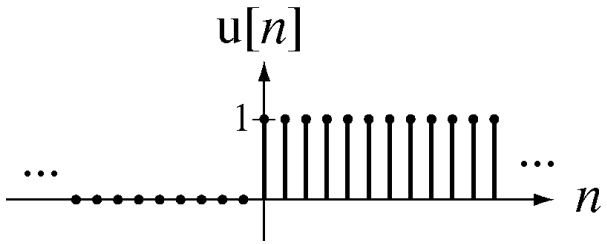
  - The z-transform of $x[n] = \delta[n-k]$ is given by:

$$X(z) = \sum_{n \in \mathbb{N}} x[n] z^{-n} = x[k] z^{-k} = z^{-k}$$

Discrete-Time singularity functions (2)

- The DT function that corresponds to the CT unit step function is called the *unit step sequence function*:

$$u[n] = \begin{cases} 1, & n \geq 0; \\ 0, & n < 0. \end{cases}$$



- The z-transform of $x[n] = u[n]$ is given by:

$$X(z) = \sum_{n \in \mathbb{N}} x[n] z^{-n} = \sum_{n \in \mathbb{N}} z^{-n}$$

- The general formula for the summation of a finite geometric series is:

$$1 + r + r^2 + r^3 + \cdots + r^{N-1} = \sum_{n=0}^{N-1} r^n = \begin{cases} N, & r = 1; \\ \frac{1-r^N}{1-r}, & r \neq 1. \end{cases}$$

Discrete-Time singularity functions (3)

- Consequently:

$$X(z) = \sum_{n \in \mathbb{N}} z^{-n} = \frac{1 - \lim_{n \to +\infty} z^{-n}}{1 - z^{-1}}$$

- The above summation will converge if $\left|z^{-1}\right| < 1$ implying that $|z| > 1$. In this case:

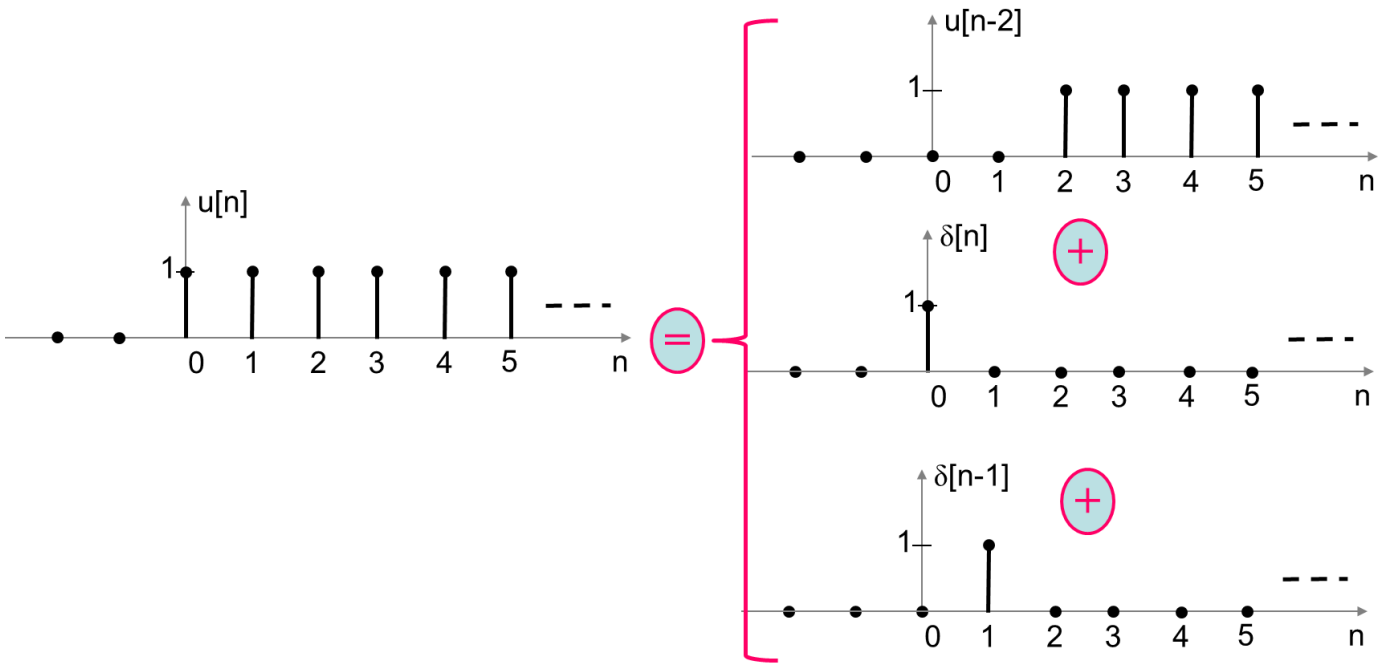
$$X(z) = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1}$$

- The Region Of Convergence is outside the unit circle $|z| = 1$.

Discrete-Time singularity functions (4)

- The unit impulse sequence and unit step sequence are related to each other.
- For example:

$$u[n] = u[n-2] + \delta[n] + \delta[n-1]$$



Discrete-Time singularity functions (5)

Multiplication by the unit impulse sequence:

$$x[n]\delta[n-k] = x[k]\delta[n-k]$$

Proof:

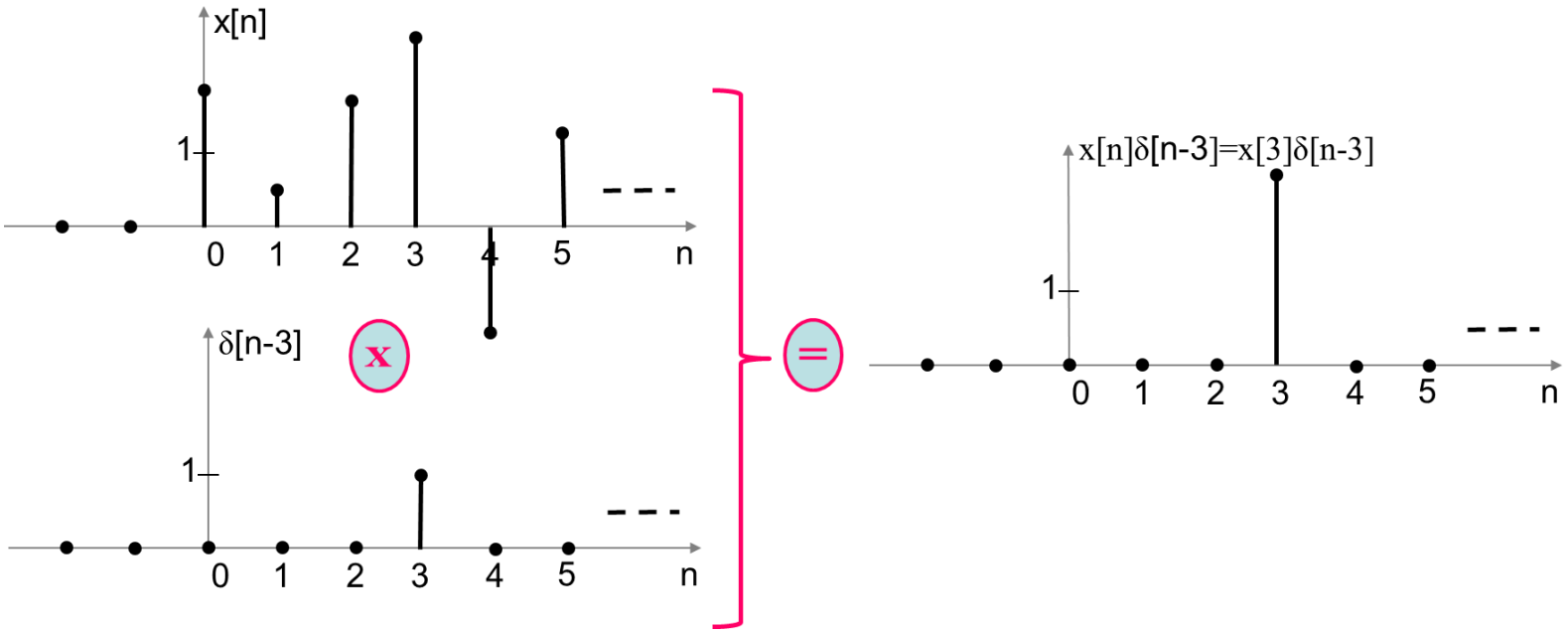
- For $n \neq k$, $\delta[n-k] = 0$ and $x[n]\delta[n-k] = 0$.
- For $n = k$, $\delta[n-k] = 1$ and $x[n]\delta[n-k] = x[k]$.
- Consequently:

$$x[n]\delta[n-k] = \begin{cases} 0, & n \neq k; \\ x[k], & n = k. \end{cases} = x[k] \underbrace{\begin{cases} 0, & n \neq k; \\ 1, & n = k. \end{cases}}_{=\delta[n-k]}$$

$$= x[k]\delta[n-k]$$

Discrete-Time singularity functions (6)

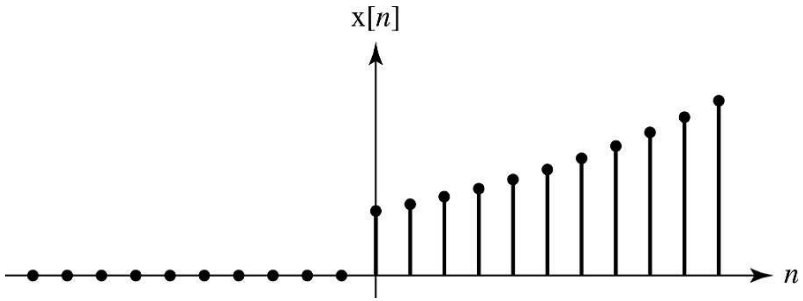
- The above property is better illustrated in the figure below (for $k = 3$):



Example-1

Calculate the z-transform of the exponential sequence:

$$x[n] = A\alpha^n u[n]$$

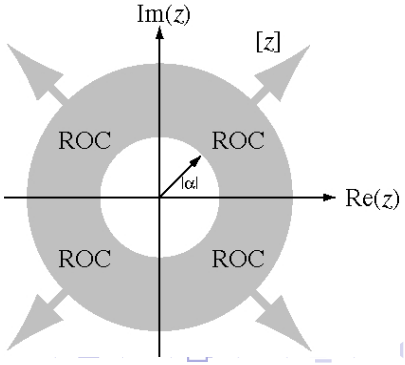


$$X(z) = \sum_{n\in\mathbb{N}} x[n]z^{-n} = A\sum_{n\in\mathbb{N}} \alpha^n z^{-n} = A\sum_{n\in\mathbb{N}} (\alpha z^{-1})^n$$

$$= A\frac{1}{1-\alpha z^{-1}} \quad \text{if} \quad \left|\alpha z^{-1}\right| < 1$$

Therefore $X(z)$ is defined by:

- $X(z) = A\frac{z}{z-\alpha}$.
- $ROC = \{z \; ; \; |z| > |\alpha|\}$.



Properties of the z-transform (1)

- Linearity:

$$\alpha x[n] + \beta y[n] \leftrightarrow \alpha X(z) + \beta Y(z)$$

- Initial value theorem:

$$x[0] = \lim_{z \to +\infty} X(z)$$

  - In fact: $X(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots$. Since $\lim_{z \to +\infty} z^{-n} = 0$ for $n > 0$, then $\lim_{z \to +\infty} X(z) = x[0]$.

- Final value theorem:

$$\lim_{n \to +\infty} x[n] = \lim_{z \to 1} (z-1)X(z)$$

if the limit exists.

Properties of the z-transform (2)

Proposition (Delay Theorem):

$$x[n] \leftrightarrow X(z)$$

$$x[n-k] \leftrightarrow z^{-k}X(z) \;\; ; \;\; k \in \mathbb{N}$$

Proof:

$$\mathcal{Z}(x[n-k]) = \sum_{n\in\mathbb{N}} x[n-k]z^{-n}$$

- Let $m = n - k$, then:

$$\mathcal{Z}(x[n-k]) = \sum_{m=-k}^{+\infty} x[m]z^{-m-k}$$

- Since $x[m] = 0$ for $m < 0$ (causal DT signal):

$$\mathcal{Z}(x[n-k]) = \sum_{m=0}^{+\infty} x[m]z^{-m-k}$$

$$= z^{-k}\sum_{m=0}^{+\infty} x[m]z^{-m} = z^{-k}X(z)$$

Properties of the z-transform (3)

Examples:

- Unit impulse:

$$\delta[n] \leftrightarrow 1$$
$$\delta[n-k] \leftrightarrow z^{-k}$$

- Exponential Sequence:

$$\alpha^n u[n] \leftrightarrow \frac{z}{z-\alpha}$$
$$\alpha^{n-k} u[n-k] \leftrightarrow \frac{z^{-k+1}}{z-\alpha}$$

- Unit Sequence:

$$u[n] \leftrightarrow \frac{z}{z-1}$$
$$u[n-k] \leftrightarrow \frac{z^{-k+1}}{z-1}$$

Properties of the z-transform (4)

Proposition (z-domain differentiation):

$$x[n] \leftrightarrow X(z)$$

$$-nx[n] \leftrightarrow z\frac{\mathrm{d}X(z)}{\mathrm{d}z}$$

Proof:

$$X(z) = \sum_{n=0}^{+\infty} x[n]z^{-n}$$

$$\Rightarrow \frac{\mathrm{d}X(z)}{\mathrm{d}z} = \sum_{n=0}^{+\infty} x[n]\left(-nz^{-n-1}\right)$$

$$\Rightarrow \frac{\mathrm{d}X(z)}{\mathrm{d}z} = -z^{-1}\sum_{n=0}^{+\infty} nx[n]z^{-n}$$

$$\Rightarrow z\frac{\mathrm{d}X(z)}{\mathrm{d}z} = \sum_{n=0}^{+\infty}(-nx[n])z^{-n} = \mathcal{Z}\left(-nx[n]\right)$$

Properties of the z-transform (5)

Example: Find the z-transform of $x[n] = n\alpha^n u[n]$.

- Using the z-domain differentiation property:

$$\alpha^n u[n] \leftrightarrow \frac{z}{z-\alpha}$$

$$\Rightarrow \quad -n\alpha^n u[n] \leftrightarrow z\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{z}{z-\alpha}\right]$$

$$\Rightarrow \quad -n\alpha^n u[n] \leftrightarrow -\frac{\alpha z}{(z-\alpha)^2}$$

$$\Rightarrow \quad n\alpha^n u[n] \leftrightarrow \frac{\alpha z}{(z-\alpha)^2}$$

Inverse z-transform (1)

In order to evaluate the inverse z-transform of $X(z)$:

- We apply the partial fraction expansion on $\frac{X(z)}{z}$.
- We apply the relation: $\mathcal{Z}(\alpha^n u[n]) = \frac{z}{z-\alpha}$.

Example 1: Calculate the inverse z-transform of:

$$X(z) = \frac{z^2}{(z-1)(z-0.2)}$$

- $\frac{X(z)}{z}$ can be written as:

$$\frac{X(z)}{z} = \frac{z}{(z-1)(z-0.2)} = \frac{A}{z-1} + \frac{B}{z-0.2}$$

where:

$$A = \left[\frac{z}{z-0.2}\right]_{z=1} = 1.25$$

$$B = \left[\frac{z}{z-1}\right]_{z=0.2} = -0.25$$

Inverse z-transform (2)

- Consequently:

$$X(z) = 1.25\frac{z}{z-1} - 0.25\frac{z}{z-0.2}$$

- Finally:

$$x[n] = \left[1.25 - 0.25(0.2)^n\right] u[n]$$

- Check the initial value theorem:
  - $x[0] = \left[1.25 - 0.25(0.2)^0\right] u[0] = 1.25 - 0.25 = 1.$
  - $\lim_{z\to+\infty} X(z) = \lim_{z\to+\infty} \frac{z^2}{(z-1)(z-0.2)} = 1.$
- Check the final value theorem:
  - $\lim_{n\to+\infty} x[n] = 1.25.$
  - $\lim_{z\to1}(z-1)X(z) = \lim_{z\to1} \frac{z^2}{z-0.2} = 1.25.$

Inverse z-transform (3)

Example 2: Calculate the inverse z-transform of:

$$Y(z) = \frac{1}{(z-1)(z-0.2)}$$

- $\frac{Y(z)}{z}$ can be written as:

$$\frac{Y(z)}{z} = \frac{1}{z(z-1)(z-0.2)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z-0.2}$$

where:

$$A = \left[\frac{1}{(z-1)(z-0.2)}\right]_{z=0} = 5$$

$$B = \left[\frac{1}{z(z-0.2)}\right]_{z=1} = 1.25$$

$$C = \left[\frac{1}{z(z-1)}\right]_{z=0.2} = -6.25$$

Inverse z-transform (4)

- Consequently:

$$Y(z) = 5 + \frac{1.25z}{z-1} - \frac{6.25z}{z-0.2}$$

- Finally:

$$y[n] = 5\delta[n] + 1.25u[n] - 6.25(0.2)^n u[n]$$

- From example 1 and example 2:

$$X(z) = \frac{z^2}{(z-1)(z-0.2)} \leftrightarrow x[n] = 1.25u[n] - 0.25(0.2)^n u[n]$$

$$Y(z) = \frac{1}{(z-1)(z-0.2)} \leftrightarrow y[n] = 5\delta[n] + 1.25u[n] - 6.25(0.2)^n u[n]$$

Inverse z-transform (5)

- Given that $Y(z) = z^{-2}X(z)$, then $y[n]$ must be equal to $x[n-2]$. Therefore, we have to check if the following quantities are equal:

$$x[n-2] = 1.25u[n-2] - 0.25(0.2)^{(n-2)}u[n-2]$$
$$y[n] = 5\delta[n] + 1.25u[n] - 6.25(0.2)^n u[n]$$

  - For $n = 0$:
    - $x[-2] = 0$ since $x[n]$ is causal.
    - $y[0] = 5\delta[0] + 1.25u[0] - 6.25(0.2)^0 u[0] = 5 + 1.25 - 6.25 = 0$.
  - For $n = 1$:
    - $x[-1] = 0$ since $x[n]$ is causal.
    - $y[1] = 5\delta[1] + 1.25u[1] - 6.25(0.2)^1 u[1] = 1.25 - 1.25 = 0$.
  - For $n \geq 2$:

$$\begin{aligned} x[n-2] &= 1.25 - 0.25(0.2)^{(n-2)} = 1.25 - 0.25(0.2)^{-2}(0.2)^n \\ &= 1.25 - 6.25(0.2)^n \\ y[n] &= 1.25 - 6.25(0.2)^n \end{aligned}$$

Inverse z-transform (6)

- The above check can be performed in a different way. Given that $u[n-2] = u[n] - \delta[n] - \delta[n-1]$, then:

$$
\begin{aligned}
x[n-2] &= \left(1.25 - 0.25(0.2)^{n-2}\right) u[n-2] \\
&= \left(1.25 - 0.25(0.2)^{n-2}\right) (u[n] - \delta[n] - \delta[n-1]) \\
&= \left(1.25 - 0.25(0.2)^{n-2}\right) u[n] \\
&- \underbrace{\left(1.25 - 0.25(0.2)^{0-2}\right)}_{=-5} \delta[n] - \underbrace{\left(1.25 - 0.25(0.2)^{1-2}\right)}_{=0} \delta[n-1]
\end{aligned}
$$

where the multiplication property of the impulse sequence was applied.

- Finally:

$$
\begin{aligned}
x[n-2] &= 5\delta[n] + \left(1.25 - 0.25(0.2)^{n}(0.2)^{-2}\right) u[n] \\
&= 5\delta[n] + \left(1.25 - 6.25(0.2)^{n}\right) u[n] = y[n]
\end{aligned}
$$

Inverse z-transform (7)

Find $x[n]$ when:

$$X(z) = \frac{z + \frac{1}{2}}{z - \frac{1}{2}}$$

First Method:

$$\frac{X(z)}{z} = \frac{z + \frac{1}{2}}{z(z - \frac{1}{2})} = \frac{A}{z} + \frac{B}{z - \frac{1}{2}}$$

- where:

$$A = \left[\frac{z + \frac{1}{2}}{z - \frac{1}{2}}\right]_{z=0} = -1 \ ; \ B = \left[\frac{z + \frac{1}{2}}{z}\right]_{z=\frac{1}{2}} = 2$$

- Consequently:

$$X(z) = -1 + 2\frac{z}{z - \frac{1}{2}}$$

$$\Rightarrow x[n] = -\delta[n] + 2\left(\frac{1}{2}\right)^n u[n]$$

Inverse z-transform (8)

Second Method:

$$X(z) = \frac{z+\frac{1}{2}}{z-\frac{1}{2}} = \frac{z}{z-\frac{1}{2}} + \frac{1}{2}\frac{1}{z-\frac{1}{2}}$$

$$= \frac{z}{z-\frac{1}{2}} + \frac{1}{2}z^{-1}\frac{z}{z-\frac{1}{2}}$$

$$\Rightarrow x[n] = \left(\frac{1}{2}\right)^n u[n] + \frac{1}{2}\left(\frac{1}{2}\right)^{n-1} u[n-1]$$

$$= \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[n-1]$$

$$= \left(\frac{1}{2}\right)^n [u[n] + u[n-1]]$$

$$= \left(\frac{1}{2}\right)^n [u[n] + (u[n] - \delta[n])]$$

$$= -\left(\frac{1}{2}\right)^n \delta[n] + 2\left(\frac{1}{2}\right)^n u[n] = -\delta[n] + 2\left(\frac{1}{2}\right)^n u[n]$$

Inverse z-transform (9)

Inversion by identifying coefficients in the power series:

$$X(z) = \sum_{n=0}^{+\infty} x[n] z^{-n}$$

Example: Calculate the inverse z-transform of:

$$X(z) = 1 + 2z^{-4} - 3z^{-5}$$

- By inspection, we deduce that:

$$\begin{aligned} x[0] &= 1 \\ x[4] &= 2 \\ x[5] &= -3 \\ x[n] &= 0 \; ; \; n \neq 0, 4, 5 \end{aligned}$$

- Note that this method can be applied only when the DT sequence has a finite duration.

Inverse z-transform (10)

- The above inspection is equivalent to the following inversion. Given that $\delta[n-k] \leftrightarrow z^{-k}$:

$$
\begin{aligned}
X(z) &= 1 + 2z^{-4} - 3z^{-5} \\
\Rightarrow \quad x[n] &= \delta[n] + 2\delta[n-4] - 3\delta[n-5] \\
&= \begin{cases} 1, & n = 0; \\ 2, & n = 4; \\ -3, & n = 5; \\ 0, & n \neq 0, 4, 5. \end{cases}
\end{aligned}
$$

Convolution-Multiplication Duality

- The convolution between two DT sequences $x[n]$ and $h[n]$ is defined as:

$$y[n] = x[n] * h[n] = \sum_{k=0}^{n} x[k]h[n-k]$$
$$= h[n] * x[n] = \sum_{k=0}^{n} h[k]x[n-k]$$

- The convolution-multiplication duality implies that:

$$x[n] * h[n] \leftrightarrow X(z)H(z)$$
$$x[n]h[n] \leftrightarrow X(z) * H(z)$$

- In other words:

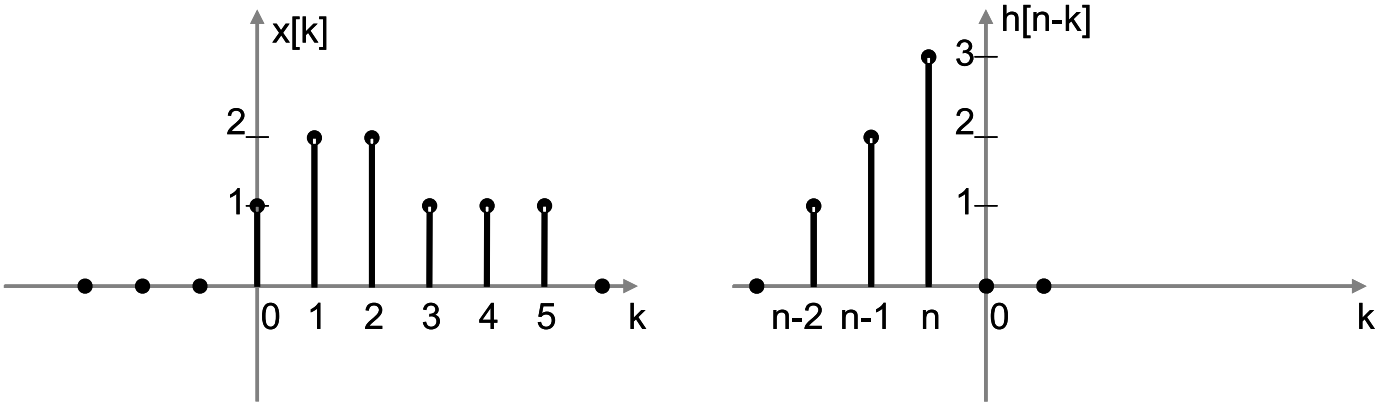
$$\mathcal{Z}(x[n] * h[n]) = \mathcal{Z}(x[n])\mathcal{Z}(h[n])$$
$$\mathcal{Z}(x[n]h[n]) = \mathcal{Z}(x[n]) * \mathcal{Z}(h[n])$$

Example-2 (1)

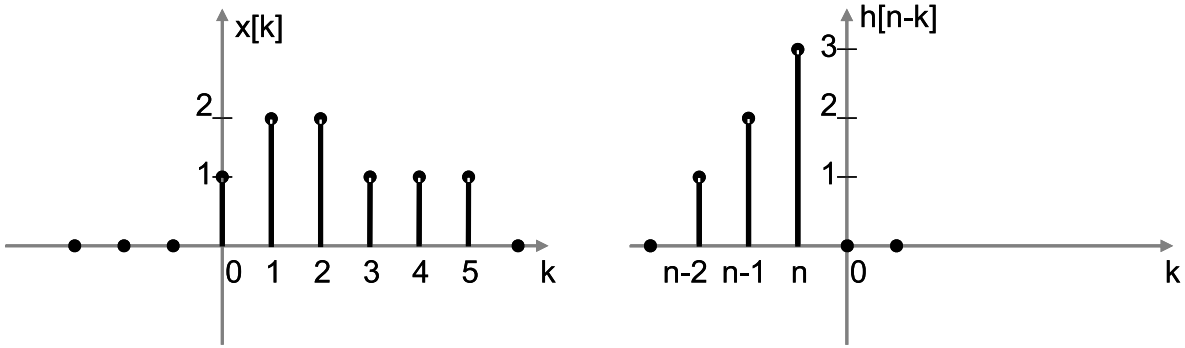
Evaluate $y[n] = x[n] * h[n]$ when $x[n]$ and $h[n]$ are given by:

- $y[n]$ is given by:

$$y[n] = \sum_{k=0}^{n} x[k]h[n-k]$$



Example-2 (2)



- Consequently:

$$y[0] = x[0]h[0] = (1)(3) = 3$$
$$y[1] = x[0]h[1] + x[1]h[0] = (1)(2) + (2)(3) = 8$$
$$y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] = (1)(1) + (2)(2) + (2)(3) = 11$$
$$\vdots$$
$$y[n] = 0 \ \text{ for } \ n < 0$$
$$y[n] = 0 \ \text{ for } \ n - 2 > 5 \quad \Rightarrow \quad n > 7$$

Example-2 (3)

Second Method:

- Since $y[n] = x[n] * h[n]$ then:

$$
\begin{aligned}
Y(z) &= X(z)H(z) \\
&= \left[1 + 2z^{-1} + 2z^{-2} + z^{-3} + z^{-4} + z^{-5}\right]\left[3 + 2z^{-1} + z^{-2}\right] \\
&= 3 + 6z^{-1} + 6z^{-2} + 3z^{-3} + 3z^{-4} + 3z^{-5} \ldots \\
&\quad + 2z^{-1} + 4z^{-2} + 4z^{-3} + 2z^{-4} + 2z^{-5} + 2z^{-6} \ldots \\
&\quad\quad + z^{-2} + 2z^{-3} + 2z^{-4} + z^{-5} + z^{-6} + z^{-7}
\end{aligned}
$$

- Finally:

$$Y(z) = 3 + 8z^{-1} + 11z^{-2} + 9z^{-3} + 7z^{-4} + 6z^{-5} + 3z^{-6} + z^{-7}$$

implying that:

$$y[n] = \{3, 8, 11, 9, 7, 6, 3, 1\}$$

Example-3 (1)

Consider the DT sequences $x_1[n]$ and $x_2[n]$ whose z-transforms are given by:

$$X_1(z) = \frac{z}{z - \frac{1}{2}} \quad ; \quad X_2(z) = \frac{z}{z - \frac{1}{4}}$$

a. Calculate $x_1[n]$ and $x_2[n]$ and deduce the value of $x[n] = x_1[n] * x_2[n]$.
b. Calculate $X(z) = X_1(z)X_2(z)$ and deduce $x[n]$.

- $x_1[n]$ is given by:

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n]$$

- $x_2[n]$ is given by:

$$x_2[n] = \left(\frac{1}{4}\right)^n u[n]$$

Example-3 (2)

- $x[n] = x_1[n] * x_2[n]$ implies that:

$$
\begin{aligned}
x[n] &= \sum_{k=0}^{n} x_1[k]x_2[n-k] \\
&= \sum_{k=0}^{n} \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} = \left(\frac{1}{4}\right)^n \sum_{k=0}^{n} \left(\frac{1}{2}/\frac{1}{4}\right)^k \\
&= \left(\frac{1}{4}\right)^n \sum_{k=0}^{n} 2^k = \left(\frac{1}{4}\right)^n \frac{1-2^{n+1}}{1-2} = \left(\frac{1}{4}\right)^n [2^{n+1}-1] \\
&= 22^n \left(\frac{1}{4}\right)^n - \left(\frac{1}{4}\right)^n \\
&= \left[2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n\right] u[n]
\end{aligned}
$$

where the last equation follows from $x[n]$ being causal (since $x_1[n]$ and $x_2[n]$ are causal).

Example-3 (3)

- $X(z)$ can be calculated from:

$$X(z) = X_1(z)X_2(z) = \frac{z^2}{(z-\frac{1}{2})(z-\frac{1}{4})}$$

- $\frac{X(z)}{z}$ can be written as:

$$\frac{X(z)}{z} = \frac{z}{(z-\frac{1}{2})(z-\frac{1}{4})} = \frac{A}{z-\frac{1}{2}} + \frac{B}{z-\frac{1}{4}}$$

where:

$$A = \left[\frac{z}{z-\frac{1}{4}}\right]_{z=1/2} = 2 \;\; ; \;\; B = \left[\frac{z}{z-\frac{1}{2}}\right]_{z=1/4} = -1$$

- Consequently:

$$X(z) = 2\frac{z}{z-\frac{1}{2}} - \frac{z}{z-\frac{1}{4}} \Rightarrow x[n] = \left[2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n\right]u[n]$$